

TAEHO HWANG

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RESEARCH INTERESTS

File system, Object Store, Non-volatile Memory, and Scalability

EDUCATION

PhD

Computer Science

03/2012 - 08/2020

Advisor: Dr. Youjip Won

Thesis: A Study on File System and Object Store for Persistent Memory

Hanyang University

MS

Electronics and Computer Engineering

03/2010 - 02/2012

Advisor: Dr. Youjip Won

Thesis: Heap-based Persistent Object Store with Byte-addressable Non-volatile Memory

Hanyang University

BS

Computer Science

03/2006 - 02/2010

Hanyang University

PUBLICATIONS

- 1 LODIC: Logical Distributed Counting for Scalable File Access**
Jeoungahn Park, **Taeho Hwang**, Jongmoo Choi, Changwoo Min, and Youjip Won
USENIX Annual Technical Conference (ATC), 2021
- 2 Copy-on-Write with Adaptive Differential Logging for Persistent Memory**
Taeho Hwang and Youjip Won
IEICE Transactions on Information and System, Vol. E102.D, Issue 12, 2019
- 3 On the I/O Characteristics of the Mobile Web Browsers**
Taeho Hwang, Myungsik Kim, Seongjin Lee, and Youjip Won
ACM Symposium on Applied Computing (SAC), 2018
- 4 Designing Persistent Heap for Byte Addressable NVRAM**
Taeho Hwang, Dokeun Lee, Yeonjin Noh, and Youjip Won
IEEE Non-Volatile Memory Systems and Applications Symposium (NVMSA), 2017
- 5 FastIO: High Speed Launching of Smart TV Apps**
Choehee Lee, **Taeho Hwang**, Youjip Won, and Seongjin Lee
Journal of KIISE (Domestic Journal in Korea), Vol. 43, No. 7, 2016
- 6 SmartCon: Smart Context Switching for Fast Block Devices**
Jongmin Gim, **Taeho Hwang**, Youjip Won, and Krishna Kant
ACM Transactions on Storage, Vol. 11, Issue 2, 2015
- 7 HEAPO: Heap-based Persistent Object Store**
Taeho Hwang, Jaemin Jung, and Youjip Won
ACM Transactions on Storage, Vol. 11, Issue 1, 2014
Github link: [HEAPO](#)

8 Improving I/O Performance in Smart TVs

Choeelhee Lee, **Taeho Hwang**, and Youjip Won

IEEE International Conference on Future Internet of Things and Cloud (FiCloud), 2014

AWARDS AND GRANTS

Student Grant Recipient	FAST'11, 02/2011
Silver Prize Recipient for Graduation Project	Hanyang University, Republic of Korea, 11/2009
Honor Student	Hanyang University, Republic of Korea, 09/2009
Honor Student	Hanyang University, Republic of Korea, 03/2009
Honor Student	Hanyang University, Republic of Korea, 03/2008
Honor Student	Hanyang University, Republic of Korea, 09/2007

WORK EXPERIENCE

Research Engineer at Bluebird 02/2014 - 02/2017

Developed payment-related security software that ranges from a library to a secure processor.

Developed a sample application for iOS, Windows CE, and Embedded Linux.

Took charge of acquiring a PCI PTS certificate.

TL at SK hynix

01/2021 -

Memory Systems Research (MSR)

TEACHING ASSISTANT EXPERIENCE

Advanced Operating System	Hanyang University, Republic of Korea, 03/2018 - 06/2018
Advanced Operating System	Hanyang University, Republic of Korea, 03/2011 - 06/2011

RESEARCH PROJECTS

GIFS: Global Index for File System on Non-volatile Memory

The virtual file system (VFS) layer uses a per-directory lock or a per-file lock for consistency. As non-volatile memory (NVM) emerges, several NVM file systems redesign file system layout. However, most NVM file systems still rely on the global lock of the VFS layer. The global lock serializes file system operations and renders file systems suffer poor scalability, especially on a shared directory or a shared file. In this project, we present Global Index File System, *GIFS*, on non-volatile memory. According to the characteristics of the NVM, we redesign an index structure for NVM file system. We replace a per-directory index and a per-file index with a global index that manages all entries in the file system. GIFS with the global index provides high concurrency and guarantees file system consistency.

Copy-on-Write with Adaptive Differential Logging for Non-volatile Memory

File systems based on non-volatile memory (NVM) deploy Copy-on-Write (COW) or logging to guarantee data consistency. However, COW has a write amplification problem and logging has a double write problem. Both COW and logging increase write traffic on NVM. In this project, we present *adaptive differential logging* and *zero-copy logging* for NVM. Adaptive differential logging applies COW or logging selectively to each block. If the updated size of a block is smaller than or equal to half of the block size, we apply logging to the block. If the updated size of a block is larger than half of the block size, we apply COW to the block. Zero-copy logging treats an user buffer on NVM as a redo log. Zero-copy logging does not incur any additional data copy.

HEAPO: Heap-based Persistent Object Store

In this project, we present a Heap-Based Persistent Object Store, *HEAPO*, to manage persistent objects in non-volatile memory (NVM). *HEAPO* defines its own persistent heap layout, the persistent object

format, name space organization, object sharing and protection mechanism, and undo-only log-based crash recovery, all of which are effectively tailored for NVM. We put our effort into developing a lightweight and flexible layer to exploit the DRAM-like access latency of NVM. To address this objective, we developed (i) a native management layer for NVM to eliminate redundancy between in-core and on-disk copies of the metadata, (ii) an expandable object format, (iii) a burst trie-based global name space with local name space caching, (iv) static address binding, and (v) minimal logging for undo-only crash recovery.